

1. Войти под пользователем user1

```
Expanded Security Maintenance for Applications is not enabled.

79 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Avksentyuk Nikita
Sun Jul 12 12:16:11 PM UTC 2026
Last login: Mon Jul 13 04:37:52 2026 from 172.16.8.4
root@eltex-practice2-v2:~# w
 09:08:01 up 5 days, 18:05,  1 user,  load average: 0.00, 0.00, 0.00
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
root      172.16.8.4    09:07          26:53m   0.00s   ?      sshd: root@pts/0
root@eltex-practice2-v2:~# su - user1
Warning: your password will expire in 5 days.
user1@eltex-practice2-v2:~$ user1
Command 'user1' not found, did you mean:
  command 'users' from deb coreutils (9.4-3ubuntu6.2)
  command 'userv' from deb userv (1.2.1-beta4)
Try: apt install <deb name>
user1@eltex-practice2-v2:~$ _
```

2. Подсчитайте количество процессов, имеющих несколько потоков выполнения

```
545 systemd+ 20 0 91028 7952 6980 S 0.0 0.2 0:01.09 systemd-timesyn
559 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-cfg80
681 root 20 0 6824 2972 2728 S 0.0 0.1 0:00.57 cron
682 message+ 20 0 9896 5716 4728 S 0.0 0.1 0:01.11 dbus-daemon
688 polkitd 20 0 383704 9908 7472 S 0.0 0.2 0:06.86 polkitd
689 root 20 0 80712 4288 3988 S 0.0 0.1 3:15.62 qemu-ga
694 root 20 0 18192 8988 7896 S 0.0 0.2 0:00.58 systemd-logind
696 root 20 0 468980 13676 11436 S 0.0 0.3 0:05.38 udisksd
713 syslog 20 0 222508 6408 4668 S 0.0 0.2 0:00.30 rsyslogd
715 root 20 0 6104 2036 1900 S 0.0 0.1 0:00.00 agetty
user1@eltex-practice2-v2:~$ ps -eL -o pid | sort | uniq -c | awk '$1 > 1 {count++} END {print count}'
10
user1@eltex-practice2-v2:~$
```

3. Запустите top и настройте вывод полей с информацией о процессе:

- удалить поля VIRT, RES, SHR;

- добавить поле RUSER и сделать чтобы это поле было после поля USER

```
Fields Management for window 1:Def, whose current sort field is %CPU
Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
'd' or <Space> toggles display, 's' sets sort. Use 'q' or <Esc> to end!

* PID      = Process Id          PSsh      = Proportion Shrd, KiB
* USER     = Effective User Name  USS        = Unique RSS, KiB
* RUSER     = Real User Name      ioR        = I/O Bytes Read
* PR        = Priority           ioRop      = I/O Read Operations
* NI        = Nice Value         ioW        = I/O Bytes Written
* VIRT      = Virtual Image (KiB) ioWop      = I/O Write Operations
* RES       = Resident Size (KiB) AGID        = Autogroup Identifier
* SHR       = Shared Memory (KiB) AGNI        = Autogroup Nice Value
* S         = Process Status     STARTED    = Start Time from boot
* %CPU      = CPU Usage          ELAPSED    = Elapsed Running Time
* %MEM      = Memory Usage (RES) %CUU       = CPU Utilization
* TIME+    = CPU Time, hundredths %CUC       = Utilization + child
* COMMAND   = Command Name/Line nsCGROUP   = CGRP namespace Inode
* PPID      = Parent Process pid nsTIME     = TIME namespace Inode
* UID       = Effective User Id
* RUID      = Real User Id
* SUID      = Saved User Id
* SUSER     = Saved User Name
* GID       = Group Id
* GROUP     = Group Name
* PGRP      = Process Group Id
```

MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3472.7 avail Mem									
PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
1	root	root	20	0	S	0.0	0.3	0:08.34	systemd
2	root	root	20	0	S	0.0	0.0	0:00.07	kthreadd
3	root	root	20	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-rcu_p
6	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-slub_
7	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
12	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-mm_pe
13	root	root	20	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
14	root	root	20	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
15	root	root	20	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
16	root	root	20	0	S	0.0	0.0	0:00.10	ksoftirqd/0

4. В другом терминальном окне выполните команду `passwd` и оставьте её в состоянии запроса текущего пароля

```
user1@eltex-practice2-v2:~$
user1@eltex-practice2-v2:~$
user1@eltex-practice2-v2:~$
user1@eltex-practice2-v2:~$
user1@eltex-practice2-v2:~$
user1@eltex-practice2-v2:~$
user1@eltex-practice2-v2:~$ passwd
Changing password for user1.
Current password:
```

5. Перейти в терминальное окно с `top` и выполнить следующие действия:

- Выведите все процессы, для которых реальным пользователем является пользователь, которым вошли в сеанс;

MiB Mem : 3915.9 total, 2836.6 free, 443.2 used, 878.0 buff/cache									
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3472.7 avail Mem									
Which user (blank for all) user1									
PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
21080	root	root	20	0	I	0.3	0.0	0:00.18	kworker/u4:1-events_unbound
21249	root	root	20	0	I	0.3	0.0	0:00.01	kworker/0:3-events
1	root	root	20	0	S	0.0	0.3	0:08.34	systemd
2	root	root	20	0	S	0.0	0.0	0:00.07	kthreadd
PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
21058	user1	user1	20	0	S	0.0	0.1	0:00.01	bash
21203	user1	user1	20	0	S	0.0	0.1	0:00.01	bash
21254	user1	user1	20	0	R	0.0	0.2	0:00.27	top

```

MiB Mem: 3915.9 total, 2841.8 free, 438.0 used, 878.1 buff
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3476.2 avail
Send pid 21251 signal [15/sigterm] 15

```

PID	USER	RUSER	PR	NI	S	%CPU	%MEM
68	root	root	0	-20	I	0.0	0.0
75	root	root	0	-20	I	0.0	0.0
78	root	root	0	-20	I	0.0	0.0
83	root	root	0	-20	I	0.0	0.0

```

21244 root root 20 0 I 0.0 0.0 0:00.03 kworker/u4:3-events_unbound
21249 root root 20 0 I 0.0 0.0 0:00.06 kworker/0:3-events
21251 root user1 20 0 S 0.0 0.1 0:00.00 passwd
21259 user1 user1 20 0 R 0.0 0.2 0:00.19 top

```

- Отправить этому процессу сигналы 2;

```

MiB Mem: 3915.9 total, 2840.1 free, 439.8 used, 878.1 buff
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3476.2 avail
Send pid 21251 signal [15/sigterm] 2

```

PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
66	root	root	0	-20	I	0.0	0.0	0:00.00	kworker

```

21244 root root 20 0 I 0.0 0.0 0:00.04 kworker/u4:3-events_power_efficient
21249 root root 20 0 I 0.0 0.0 0:00.08 kworker/0:3-events
21251 root user1 20 0 S 0.0 0.1 0:00.00 passwd
21263 root root 20 0 I 0.0 0.0 0:00.00 kworker/0:0
21264 root root 20 0 I 0.0 0.0 0:00.00 kworker/0:2-cgroup_bpf_destroy
21265 root root 20 0 I 0.0 0.0 0:00.01 kworker/u4:2-events_power_efficient
21267 user1 user1 20 0 R 0.0 0.2 0:00.15 top

```

- Отправить этому процессу сигналы 3;

```

%cpu(s): 0.0 us, 0.2 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem: 3915.9 total, 2841.8 free, 438.0 used, 878.1 buff
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3477.9 avail
Send pid 21251 signal [15/sigterm] 3

```

PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
93	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-c
156	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-t
171	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-k

```

21251 root user1 20 0 S 0.0 0.1 0:00.00 passwd
21263 root root 20 0 I 0.0 0.0 0:00.00 kworker/0:0
21264 root root 20 0 I 0.0 0.0 0:00.00 kworker/0:2-cgroup_bpf_destroy
21265 root root 20 0 I 0.0 0.0 0:00.02 kworker/u4:2-events_unbound
21269 user1 user1 20 0 R 0.0 0.2 0:00.13 top

```

- Отправить этому процессу сигналы 9;

```

top - 10:14:51 up 5 days, 19:12, 2 users, load average:
Tasks: 116 total, 2 running, 114 sleeping, 0 stopped,
%Cpu(s): 0.0 us, 0.9 sy, 0.0 ni, 99.1 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem: 3915.9 total, 2839.8 free, 439.9 used,
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used.
Send pid 21251 signal [15/sigterm] 9

```

PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+
171	root	root	0	-20	I	0.0	0.0	0:00.00
201	root	root	0	-20	I	0.0	0.0	0:00.00
240	root	root	20	0	S	0.0	0.0	0:01.34
241	root	root	0	-20	I	0.0	0.0	0:00.00
308	root	root	19	-1	S	0.0	0.7	0:02.23

```
21223 root      root      20    0 I   0.0   0.0   0:00.00 kworker/1:3
21244 root      root      20    0 I   0.0   0.0   0:00.06 kworker/u4:3-events_power_efficient
21249 root      root      20    0 I   0.0   0.0   0:00.10 kworker/0:3-events
21263 root      root      20    0 I   0.0   0.0   0:00.00 kworker/0:0
21264 root      root      20    0 I   0.0   0.0   0:00.00 kworker/0:2-cgroup_bpf_destroy
21265 root      root      20    0 I   0.0   0.0   0:00.02 kworker/u4:2-events_power_efficient
21275 user1     user1     20    0 R   0.0   0.2   0:00.14 top

user1@eltex-practice2-v2:~$
user1@eltex-practice2-v2:~$ passwd
Changing password for user1.
Current password: Killed
user1@eltex-practice2-v2:~$ _
```

6. Выполнить команду `vim ~/file_task3.txt` и нажать `Ctrl-Z`

```
~/file_task3.txt" [New]

[1]+  Stopped                  vim ~/file_task3.txt
user1@eltex-practice2-v2:~$ _
```

7. Выполнить команду `sleep 600`, нажмите `Ctrl-Z` и выполните команду `jobs`

```
~/file_task3.txt" [New]

[1]+  Stopped                  vim ~/file_task3.txt
user1@eltex-practice2-v2:~$ sleep 600
^Z
[2]+  Stopped                  sleep 600
user1@eltex-practice2-v2:~$ _

^Z
[2]+  Stopped                  sleep 600
user1@eltex-practice2-v2:~$ jobs
[1]-  Stopped                  vim ~/file_task3.txt
[2]+  Stopped                  sleep 600
user1@eltex-practice2-v2:~$
```

8. Последнее задание (`sleep 600`) сделайте фоновым

```

user1@eltex-practice2-v2:~$ jobs
[1]-  Stopped                  vim ~/file_task3.txt
[2]+  Stopped                  sleep 600
user1@eltex-practice2-v2:~$ %2
sleep 600
jobs
^Z
[2]+  Stopped                  sleep 600
user1@eltex-practice2-v2:~$ jobs
[1]-  Stopped                  vim ~/file_task3.txt
[2]+  Stopped                  sleep 600
user1@eltex-practice2-v2:~$ bg %2
[2]+ sleep 600 &
user1@eltex-practice2-v2:~$ jobs
[1]+  Stopped                  vim ~/file_task3.txt
[2]-  Running                  sleep 600 &
user1@eltex-practice2-v2:~$

```

9. Измените число NICE у задания (600), сделав его равным 10

```

user1@eltex-practice2-v2:~$ jobs -l
[1]+ 21280 Stopped              vim ~/file_task3.txt
[2]- 21286 Running              sleep 600 &
user1@eltex-practice2-v2:~$ renice -n 10 -p 21286
21286 (process ID) old priority 0, new priority 10
user1@eltex-practice2-v2:~$ jobs -l

```

10. Проверить, что число NICE у этого задания изменилось

```

[2]-  Done                    sleep 600
user1@eltex-practice2-v2:~$ jobs -l
[1]+ 21280 Stopped              vim ~/file_task3.txt
user1@eltex-practice2-v2:~$ sleep 600
^Z
[2]+  Stopped                  sleep 600
user1@eltex-practice2-v2:~$ jobs -l
[1]- 21280 Stopped              vim ~/file_task3.txt
[2]+ 21300 Stopped              sleep 600
user1@eltex-practice2-v2:~$ bg %2
[2]+ sleep 600 &
user1@eltex-practice2-v2:~$ jobs
[1]+  Stopped                  vim ~/file_task3.txt
[2]-  Running                  sleep 600 &
user1@eltex-practice2-v2:~$ renice -n 10 -p 21286
renice: failed to get priority for 21286 (process ID): No such process
user1@eltex-practice2-v2:~$ renice -n 10 -p 21300
21300 (process ID) old priority 0, new priority 10
user1@eltex-practice2-v2:~$ ps -o pid,nice,comm -p 21300
    PID  NI  COMMAND
    21300  10  sleep
user1@eltex-practice2-v2:~$ _

```

11. Сделайте задание vim ~/file_task3.txt активным и выйдите из редактора

